

Vol 2 Chapter 7 — CKM Mixing and the Jarlskog Invariant

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Chapter 7 — CKM Mixing and the Jarlskog Invariant

Legacy chapter (May 2026), corrected in place 2026-09-11 (Keeper, K1892) at the sentences the presentation scan flagged; the remaining tiers are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

What changed since this chapter was written (2026-09-11, Keeper). The mixing sector closed on 2026-08-22 (Rounds 47–61, K1791–K1810), and this chapter predates the closure. The current state: derived — the skeleton (T2519), the Cabibbo angle $\lambda = 1/\sqrt{20}$ obtained blind on 2026-07-29 (T2530), the existence of CP violation (T2547), flavour universality as the partial-isometry condition, and the order of the 1–3 corner — $|V_{ub}|$ exactly one power of λ below $|V_{cb}|$, a zero-knob statement (K1808/K1810). Input — the corner's value, δ_{CP} 's magnitude, and $|V_{cb}|$ (the old 0.044 retired at $+9.2\sigma$ against current exclusive data; the *position* kept). Fired and lost — five named series for the corner value, sealed by hash before scoring, all missed. The $2/\sqrt{79}$ form of the Cabibbo angle that T1444 supplied is retired in favour of the banked $1/\sqrt{20}$; the Jarlskog match below remains conditional as stated. The live falsifier is first-row unitarity: $1/\sqrt{20}$ with $|V_{ud}| = \sqrt{19/20}$ is exactly unitary, and the direct determinations sum 2.3σ short (PDG 2024). Spine Lecture 7 is the current treatment.

The Cabibbo–Kobayashi–Maskawa (CKM) matrix encodes quark flavor mixing in weak interactions. Three mixing angles and one CP-violating phase parameterize the matrix; the Jarlskog invariant J — the unique CP-violating phase-invariant combination — measures the strength of CP violation in the quark sector and is experimentally measured at $J \approx 3.18 \times 10^{-5}$.

BST derives the Jarlskog invariant to 0.3% match (CONDITIONAL on the T1444 vacuum-subtraction theorem; the naive substrate prediction is $\sim 3\%$ off, with the 0.3% match emerging once the vacuum-subtraction correction is applied).

7.1 The Jarlskog invariant

The Jarlskog invariant in standard parameterization:

$$J = \text{Im}(V_{us}V_{cb}V_{ub}^*V_{cs}^*) \approx 3.18 \times 10^{-5}.$$

It is the unique re-phasing-invariant combination of CKM matrix elements that measures CP violation.

7.2 The substrate derivation

The BST derivation of J runs through the substrate's Q^5 Chern-flux structure (Lyra T1920) combined with the rank-2 quark doublet structure of weak isospin. The substrate-mechanism reading: the CP-violating phase is the substrate's twist between the three quark generations on Q^5 , with the magnitude controlled by Casimir-weighted ratios of the BST primaries.

The naive substrate prediction gives J at $\sim 3\%$ from the measured value — close, but above D-tier. The T1444 vacuum-subtraction theorem (Lyra) closes the residual: the substrate's vacuum-state Casimir contribution must be subtracted from the naive evaluation, bringing the match to 0.3%.

The result is D-tier conditional on T1444 vacuum-subtraction. Without the vacuum-subtraction the result is I-tier at $\sim 3\%$ match.

7.3 The Cabibbo angle and other mixing parameters

The Cabibbo angle $\theta_C \approx 13.04^\circ$ ($\sin \theta_C \approx 0.225$) is substrate-derived via Casey's W-17 work (May 2026): the Cabibbo angle is the substrate's natural T^2 -rotation angle between up-type and down-type quark K-types in the first generation.

The other CKM mixing angles ($\theta_{12}, \theta_{23}, \theta_{13}$) are similarly substrate-derived from K-type rotational structure across generations. Per-angle match precisions span 1–5% (I-tier multi-month research target for D-tier promotion).

7.4 The decay-rate battery

The framework's decay-rate predictions span kaon mixing (Δm_K), B-meson mixing ($\Delta m_d, \Delta m_s, B_s$), and related quantities. The substrate-mechanism is the same Q^5 Chern-flux structure that gives the Jarlskog invariant; per-quantity match precisions are at the percent level. Lyra T1920 cascade.

Specific results: - Δm_K (neutral kaon mass difference): substrate-derived at $\sim 1\%$ - $\Delta m_d, \Delta m_s$ (B meson mixing): substrate-derived at $\sim 1\text{--}3\%$ - ϵ_K (kaon CP violation parameter): D-tier via T2132 VSC chain

7.5 What comes next

Chapter 8 — Coupling Constants and Anomalous Moments — treats the electron’s anomalous magnetic moment a_e at parts-per-trillion precision, the volume’s second Crown Jewel after Chapter 6’s m_p/m_e .

Where to look this up: Jarlskog J : T1444 (vacuum-subtraction theorem). Cabibbo angle: Casey W-17. Decay-rate battery: T1920 Chern-flux cascade. For standard CKM treatments: Peskin–Schroeder Chapter 20.3; Branco, Lavoura, and Silva, *CP Violation*.